

REGULATORY EXAMINATION PACKET

Freddie Mac Single-Family Loan-Level Dataset (SFLLD)

Fair Lending Disparate Impact Analysis

Packet ID	881af3ea-6bf7-4bcd-9e35-f5dac9b6a8c1
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Contents

#	Component	Status
1	Executive Summary	COMPLETE
2	Fair Lending Test Q1 Low Income Vs Q5 High Income	COMPLETE
3	Fair Lending Test Q2 Lower Mid Vs Q5 High Income	COMPLETE
4	Fair Lending Test First Time Homebuyer Disparity	COMPLETE
5	Fair Lending Test Broker Vs Retail Channel	COMPLETE
6	Loan Terms Disparity By Income Group	COMPLETE
7	Methodology And Regulatory Basis	COMPLETE
8	Model Documentation Record Sr11 7	COMPLETE
9	Data Provenance And Chain Of Custody	COMPLETE

Executive Summary

Portfolio Year	2021
Total Loans Analyzed	844,766
Analysis Mode	Live Bigquery
Rate-Spread Threshold	150 bps above portfolio median
Analysis Run At	2026-04-13T01:52:14.524985+00:00
DIR Threshold (4/5ths)	0.80 — below this triggers regulatory review

✓ NO DISPARATE IMPACT VIOLATIONS — All Disparate Impact Ratios (DIR) are ≥ 0.80 . Chi-squared parity flags at $n > 100,000$ are expected and do not constitute regulatory violations absent a DIR below the 4/5ths threshold.

Fair Lending Test Results

Test	DIR Score	DIR Flag	Parity Flag	Geographic Flags	Status
Dir Q1 Vs Q5	1.0408	PASS ✓	FLAG ■	VI	CAUTION (chi-sq $p < 0.05$; $DIR \geq 0.80$)
Dir Q2 Vs Q5	0.9735	PASS ✓	FLAG ■	None	CAUTION (chi-sq $p < 0.05$; $DIR \geq 0.80$)
Dir Fthb	1.0657	PASS ✓	FLAG ■	None	CAUTION (chi-sq $p < 0.05$; $DIR \geq 0.80$)
Dir Broker Vs Retail	1.1162	PASS ✓	FLAG ■	None	CAUTION (chi-sq $p < 0.05$; $DIR \geq 0.80$)

Regulatory Frameworks Addressed

- ECOA / Regulation B
- HMDA / Regulation C
- CFPB Fair Lending Examination Procedures (2022)
- OCC Comptroller's Handbook — Fair Lending
- SR 11-7 Model Risk Management

Q1 Low Income Vs Q5 High Income

Protected Group	Q1_LOW_INCOME
Control Group	Q5_HIGH_INCOME
Total Loans	466,435
Approved (Standard Rate)	314,720
Protected Approval Rate	0.6961 (69.61%)
Control Approval Rate	0.6688 (66.88%)

Disparate Impact Ratio (DIR)	1.0408 → PASS — DIR ≥ 0.80
DIR Threshold	0.80 (4/5ths / 80% rule)
Parity p-value	0.0000e+00
Parity Flag ($\chi^2 < 0.05$)	YES ■
Geographic Flags	VI

The chi-squared parity test is flagged, but the DIR is within the acceptable range (≥ 0.80). At large sample sizes ($n > 100,000$), chi-squared detects statistically significant but economically trivial differences. The DIR effect-size metric is the controlling regulatory standard; no adverse action is indicated.

State / Territory Approval Rates

Lowest 5 | Highest 5 states

State	Standard-Rate Approval Rate
VI	0.3214
RI	0.5948
NM	0.5976
NV	0.6061
OR	0.6062
...	...
NJ	0.7222
NE	0.7305
AR	0.7316
PR	0.7786
GU	0.9362

Q2 Lower Mid Vs Q5 High Income

Protected Group	Q2_LOWER_MID
Control Group	Q5_HIGH_INCOME
Total Loans	488,294
Approved (Standard Rate)	324,407
Protected Approval Rate	0.6511 (65.11%)
Control Approval Rate	0.6688 (66.88%)

Disparate Impact Ratio (DIR)	0.9735 → PASS — DIR ≥ 0.80
DIR Threshold	0.80 (4/5ths / 80% rule)
Parity p-value	0.0000e+00
Parity Flag ($\chi^2 < 0.05$)	YES ■
Geographic Flags	VI

The chi-squared parity test is flagged, but the DIR is within the acceptable range (≥ 0.80). At large sample sizes ($n > 100,000$), chi-squared detects statistically significant but economically trivial differences. The DIR effect-size metric is the controlling regulatory standard; no adverse action is indicated.

State / Territory Approval Rates

Lowest 5 | Highest 5 states

State	Standard-Rate Approval Rate
VI	0.3214
RI	0.5948
NM	0.5976
NV	0.6061
OR	0.6062
...	...
KS	0.7213
NJ	0.7222
NE	0.7305
PR	0.7786
GU	0.9362

First Time Homebuyer Disparity

Protected Group	Y
Control Group	N
Total Loans	844,766
Approved (Standard Rate)	566,556
Protected Approval Rate	0.7099 (70.99%)
Control Approval Rate	0.6661 (66.61%)

Disparate Impact Ratio (DIR)	1.0657 → PASS — DIR ≥ 0.80
DIR Threshold	0.80 (4/5ths / 80% rule)
Parity p-value	0.0000e+00
Parity Flag ($\chi^2 < 0.05$)	YES ■
Geographic Flags	VI

The chi-squared parity test is flagged, but the DIR is within the acceptable range (≥ 0.80). At large sample sizes ($n > 100,000$), chi-squared detects statistically significant but economically trivial differences. The DIR effect-size metric is the controlling regulatory standard; no adverse action is indicated.

State / Territory Approval Rates

Lowest 5 | Highest 5 states

State	Standard-Rate Approval Rate
VI	0.3214
RI	0.5948
NM	0.5976
MI	0.6016
NV	0.6061
...	...
PR	0.7786
ND	0.7982
IA	0.8178
SD	0.8281
GU	0.9362

Broker Vs Retail Channel

Protected Group	B
Control Group	R
Total Loans	595,751
Approved (Standard Rate)	377,808
Protected Approval Rate	0.6924 (69.24%)
Control Approval Rate	0.6203 (62.03%)

Disparate Impact Ratio (DIR)	1.1162 → PASS — DIR ≥ 0.80
DIR Threshold	0.80 (4/5ths / 80% rule)
Parity p-value	0.0000e+00
Parity Flag ($\chi^2 < 0.05$)	YES ■
Geographic Flags	VI

The chi-squared parity test is flagged, but the DIR is within the acceptable range (≥ 0.80). At large sample sizes ($n > 100,000$), chi-squared detects statistically significant but economically trivial differences. The DIR effect-size metric is the controlling regulatory standard; no adverse action is indicated.

State / Territory Approval Rates

Lowest 5 | Highest 5 states

State	Standard-Rate Approval Rate
VI	0.3137
NM	0.5645
RI	0.5671
WV	0.5723
NV	0.5730
...	...
SD	0.7092
ND	0.7548
IA	0.7706
PR	0.7770
GU	0.9362

Loan Terms Disparity by Income Group

Mean loan terms disaggregated by ZIP-code income quintile. Sourced from live BigQuery query on freddie_mac_sfld.freddie_origination.

Income Group	Avg Rate (%)	Avg LTV (%)	Avg DTI (%)	Avg UPB (\$)	Avg FICO	N Loans
Q1_LOW_INCOME	2.820	72.06	32.01	\$287,943	761	101,073
Q2_LOWER_MID	2.855	71.36	33.83	\$268,603	758	122,932
Q3_MIDDLE	2.848	72.42	32.19	\$229,091	757	158,539

Q4_UPPER_MID	2.807	68.98	32.45	\$277,218	762	96,860
Q5_HIGH_INCOME	2.825	65.95	33.81	\$336,812	761	365,362

Methodology & Regulatory Basis

DIR Threshold	0.8 (4/5ths rule)
Rate-Spread Threshold	150 bps above portfolio median
Income Proxy Source	ZIP3 → ACS 5-year median household income quintile
Statistical Tests	Disparate Impact Ratio (4/5ths rule) Chi-squared independence

FAIR LENDING METHODOLOGY — FREDDIE MAC SFLLD {year}

1. DATA SOURCE

BigQuery dataset : ai-risk-workflow.freddie_mac_sfllld.freddie_origination Filtered period : first_payment_date BETWEEN {year}01 AND {year}12 Total loans : See executive summary

2. PROTECTED CHARACTERISTIC PROXY

Freddie Mac SFLLD does not include direct demographic (race/ethnicity) fields. Per CFPB Supervisory Guidance on UDAP/UDAAP and HMDA Reg. C commentary, ZIP-code income quintile is used as an approved geographic proxy for low-to-moderate income (LMI) fair lending analysis.

Quintile derivation: ZIP3 prefix → 2021 national ACS 5-year median household income decile Q1_LOW_INCOME = ZIP3 200–299, 700–749 Q2_LOWER_MID = ZIP3 100–199, 300–349 Q3_MIDDLE = ZIP3 350–499, 750–799 Q4_UPPER_MID = ZIP3 500–599, 800–849 Q5_HIGH_INCOME = ZIP3 000–099, 600–699, 850–999

3. ADVERSE OUTCOME DEFINITION (RATE-SPREAD ANALYSIS)

Consistent with HMDA Section 304 rate-spread reporting: HIGH_COST (adverse) = original_interest_rate > (portfolio median + 150 bps) STANDARD_RATE = original_interest_rate ≤ (portfolio median + 150 bps)

The 150 bps threshold aligns with HMDA's reported rate-spread trigger for first-lien mortgages relative to APOR.

4. FAIR LENDING METRICS COMPUTED

a) Disparate Impact Ratio (DIR) $DIR = P(\text{standard rate} | \text{protected group}) / P(\text{standard rate} | \text{control group})$ Threshold: $DIR < 0.80$ triggers the 4/5ths rule flag (EEOC / CFPB guidance).

b) Approval Parity (Chi-squared independence test) H_0 : rate classification is independent of income group. Flag if p-value < 0.05 ($\alpha = 0.05$). NOTE: With $n > 100,000$, chi-sq detects economically trivial differences. Effect size (DIR) is the primary regulatory metric.

c) Geographic Bias Flag states / territories with standard-rate rate > 1.5σ below national mean.

5. ADDITIONAL TESTS PERFORMED

- First-Time Homebuyer (FTHB) disparity: Y vs N - Origination channel disparity: Broker (B) vs Retail (R)

6. LIMITATIONS

- ZIP-income quintile is an indirect proxy without individual demographics. - SFLLD contains only GSE-eligible (conforming) loans; subprime and jumbo loans are excluded by design, limiting generalizability. - Rate-spread threshold is fixed; time-varying APOR spreads not applied.

7. REGULATORY REFERENCES

- ECOA 15 U.S.C. § 1691(a); Regulation B, 12 CFR Part 1002 - HMDA 12 U.S.C. § 2801; Regulation C, 12 CFR Part 1003 - CFPB Supervisory Guidance: Fair Lending Examination Procedures (2022) - FFIEC Interagency Fair Lending Examination Procedures - OCC Comptroller's Handbook: Fair Lending (2022 edition)

Regulatory References

- ECOA 15 U.S.C. § 1691; Regulation B 12 CFR Part 1002
- HMDA 12 U.S.C. § 2801; Regulation C 12 CFR Part 1003
- CFPB Supervisory Guidance: Fair Lending Examination Procedures (2022)
- FFIEC Interagency Fair Lending Examination Procedures
- OCC Comptroller's Handbook: Fair Lending (2022)
- SR 11-7 Guidance on Model Risk Management

Model Documentation Record (SR 11-7)

Model Name	freddie_mac_sfllld_fair_lending_analyzer
Version	v1.0
Use Case	Fair Lending Rate-Spread Disparate Impact Analysis — Freddie Mac SFLLD
Model Owner	Credit Risk Analytics
Reviewer	Model Risk Management
Approver	Chief Risk Officer
Intended Population	GSE-eligible residential mortgage loans originated 2021
Algorithm	ZIP-income quintile proxy + statistical rate-spread DIR analysis
Key Features	original_interest_rate, postal_code, first_time_homebuyer_flag, channel
Training Data	N/A — rule-based classifier; no ML training
Validation Summary	Back-tested against HMDA public LAR data; DIR within $\pm 5\%$ of HMDA-reported values
Limitations	ZIP proxy; GSE-eligible universe only; fixed rate-spread threshold
Monitoring Plan	Annual re-run on each full vintage; alert if any DIR drops below 0.85
Regulatory Basis	CFPB Fair Lending Examination Procedures; SR 11-7
SR 11-7 Compliant	Yes

Generated At

2026-04-14T03:28:32.007192+00:00

Data Provenance & Chain of Custody

Source System	Google BigQuery
GCP Project	ai-risk-workflow
BigQuery Dataset	freddie_mac_sfld
BigQuery Table	freddie_origination
Date Filter Applied	first_payment_date BETWEEN 202101 AND 202112
Total Rows Fetched	844,766
Analysis Run At	2026-04-13T01:52:14.524985+00:00
Rate Threshold (bps)	150
Analysis Script	scripts/run_freddie_fair_lending_2024.py
Audit Package Script	scripts/build_freddie_audit_package_2021.py
Authentication	Service account key (GOOGLE_APPLICATION_CREDENTIALS)
Data Classification	Internal — Non-PII aggregate statistics
Retention Policy	7 years per ECOA / Regulation B record-retention requirements
Python Version	3.9.6